Raghav Pant

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EDUCATION

University of Texas at Austin, *M.S. Computational Science*, 3.91/4.0 **Austin, TX | Sep 2023 – May 2025 (exp)**
Select Courses**:** Functional Analysis, Numerical Lin Alg, Foundations of Data/ML, Stochastic Processes, Comp Linguistics + NLP

Brown University, *B.S. Applied Mathematics*, 3.9/4.0 **Providence, RI | Sep 2018 – May 2022**
Select Courses**:** Comp Stat, Machine Learning, Deep Learning, Graphs/Networks, Data Struc+Algos, PDEs, Stochastic Diff Eqs

TECHNICAL SKILLS

Data/Machine Learning: Python (numpy, scipy, pandas, scikit-learn, Pytorch, Tensorflow), HuggingFace, R, SQL (basic)
HPC/Developer Tools: C++ (basic), CUDA (basic), Docker, Git/GitHub Actions

WORK EXPERIENCE

Capstone Investment Advisors, *Quantitative Research Intern* **New York, NY | Jun 2024 – Aug 2024**
- Equity Volatility Relative Value: identified optimal systematic hedges for vega/skew strategies using historical PNL and Greeks in Python; applying deep learning methods to extract implied quantities from American option surfaces

JPMorgan Chase, *Equity Derivatives Trader* **New York, NY | Jun 2022 – Jul 2023**
- Trading: made markets and managed risk on SPX, VIX, and variance swaps (Index Flow Volatility Trading)
  - Quoted listed and OTC up to 500k RT-vega for various hedge fund and asset manager clients; responsible for managing vega positioning and VIX-SPX spreads as well as delta risk intraday
  - Helped manage index variance relative value and VIX futures rolldown books
- Research/Development: implemented Python-based projects to streamline desk operations
  - Built prototype automated rho hedging tool for index and single stock vol desks using risk data feed
  - Constructed automated dashboard to monitor variance swap and forward variance positions

Goldman Sachs, *Sales and Trading Intern* **New York, NY | Jun 2021 – Aug 2021**
- Electronic Market Making: worked with 100GB+ datasets in pandas to research off-exchange traded volume and quantify patterns in execution flows, as part of signal research and development
- Equity Derivatives Trading: built analytical (finite-difference) pricing tool for an option on a custom basket underlier

RESEARCH EXPERIENCE

Cockrell School of Engineering, UT Austin, *Graduate Research Assistant* **Austin, TX | Sep 2023 – Present**
- Iterate on graph neural network (GNN)-based architecture for data-driven physics simulation and sciML:
  - Investigating properties of Kolmogorov-Arnold Networks for improved inference and runtime in PDE-based operator learning and GNN-based simulation problems (work accepted as conference presentation at SIAM-CSE25)
  - Build multifidelity Monte Carlo techniques to reduce estimator variance and optimize training/evaluation cost tradeoff for GNN model, with applications to general neural network optimization

Dept of Applied Mathematics, Brown University, *Undergraduate Researcher* **Providence, RI | Jan 2020 – May 2022**
- Constructed ODE-based model of RSV transmission in daycare settings, incorporating novel environment-based pathways
- Completed rigorous stability analyses in Numpy/MATLAB, including PRCC on model params and R0 sensitivity

SELECTED PROJECTS

Multilingual Language Models and Embeddings *(Python/Pytorch/HuggingFace):* a research project investigating the embedding properties of SOTA multilingual LLMs for South Asian languages like Hindi/Urdu, and the impact of finetuning, chain-of-thought prompting, and related techniques to build and improve open-source LLM inference in those languages.

Song Similarity via Convolutional Neural Nets *(Pytorch):* CNNs and Siamese neural networks applied to mel spectrograms to learn/measure song similarity from audio samples (GTZAN dataset and custom song data pulled from Spotify API).

ACTIVITIES & INTERESTS

SIAM @ UT Austin; Texas Venture Labs; Captain, Brown Men's Club Soccer (2022)
Languages: English, Spanish, Hindi/Urdu; **Interests:** Soccer, hiking, Urdu poetry, linguistics